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Automatic question generation and answer assessment for subjective examination

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ABSTRACT

Automatic generation of questions and evaluating their answers is a highly challenging task in natural language processing and educational technology. This work focuses on generating subjective questions and also an evaluation system is suggested for assessing the answers. For generating the questionnaires, key-phrases are extracted from the course curriculum (syllabus). Next, based on the key-phrases, different types of subjective questions are generated. Finally, the evaluation of student's responses is achieved using a multi-criteria-decision-making approach. It uses a set of model answers taken from different textbooks and subject experts to evaluate the answers. Multiple measures are used to assess the answers by comparing them with this model set. The results of the profound system reveal that the automated appraisal process can reduce the manual effort of the human.

1. Introduction

A question is defined as a sentence that seeks an answer for the purpose of information retrieval and is widely used as a medium of assessment in education. Questions are mainly categorized into two forms: objective question and subjective question. The objective-question asks students to choose the correct answer from several alternatives or to provide a word or short phrase to answer a question or to complete a statement. Objective questions include multiple-choice, matching, true-false and fill-in-the-blank questions (Boyd, 1988), whereas the subjective question requires an answer in the form of explanation. It permits the students to organize and present responses in their own words. The subjective questions include definitions, short answer type questions, essay-type questions, scenario questions, and opinion questions (Clay, 2001). Generally, a good classroom test should contain both subjective and objective questions (Heaton, 1990). The subjective-question develops the student's critical reasoning and writing skills (Dhokrat, Gite, & Mahender, 2012). It is time-consuming and laborious to generate handcraft questions. Therefore, over the past two decades, researchers have devoted significant attention to the automatic question generation. Several automated or semi-automated systems are found in the literature for generating questions (Divate & Salgaonkar, 2017; Liu, Rus, & Liu, 2017; Soleymanzadeh, 2017). Objective question is the most popular assessment tool in the computerized examination for the last few years due to its fast, reliable, and uniform evaluation strategies (Nicol, 2007). It involves the binary-mode of assessment, which has only one correct answer. On the other hand, the subjective examination achieves a trend for evaluating a candidate's knowledge and understanding of the traditional education system for centuries (Shaban, 2014). Each university has its unique examination pattern based on subjective examination. So, when the Information and Communications Technology (ICT)-based teaching–learning methods are growing day by day, we need to consider such assessment and evaluation done by the computer-assisted appraisal system. The computerized answer-assessment of subjective-question is difficult, and the accuracy of computer-based evaluation has not achieved adequate results. Towards research on automatic assessment of subjective questions, hopefully, discovering new tools to support schools and teachers. It solves the problem of high cost and slow turnaround of hand-scoring thousands of written answers in the subjective examination. Due to the unreliable accuracy of evaluation, today's computerized-examination has excluded the subjective-questions in favor of multiple-choice questions, which are unable to assess student's writing skills and critical reasoning.

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In this paper, we have proposed a computerized system for Question Generation and Answer Assessment (QGAA) that is capable of grading student's written responses. The system attempts to generate subjective questions from a collection of course curriculums and also evaluates the answers written by the examinees. The proposed system contributes to the literature in several ways, as mentioned below.

- An automated system QGAA is proposed for subjective examination.
- The system generates subjective questions from a set of course curriculums that are helpful to evaluate the subjective-knowledge of candidates.
- We have proposed a Multi-criteria Decision Making (MCDM)-based approach for fusing the multiple similarity measures to evaluate the correctness of subjective answers.

The rest of the paper is organized as follows. Section 2 describes the related works. In Section 3, we discuss the proposed framework for generating subjective questions and assessment of answers, including an overview of the similarity measure and fusion technique. Section 4 shows the results and case studies using an in-house dataset called DB-SYLLABI. Finally, in Section 6, we conclude our paper by highlighting some future extensions of the present work.

2. Related work

To pursue the urge of modern ICT-based education, several question generation models are studied in the literature to make the teaching–learning process more effective (Katsuyama, Ozawa, & Naoi, 2010; Mitkov, Lean, & Karamanis, 2006; Nanjiani, Kuhlke, Mishra, & Ramakrishnananda, 2015; Thomas, 1999). Wolfe (1976) developed the first automatic question generation system. A sequence of developments is found in Aist (2001), Brown, Frishkoff, and Eskenazi (2005), Chen, Liou, and Chang (2006), Cheng, Lin, and Huang (2009), Kunichika, Katayama, Hirashima, and Takeuchi (2001) and Mostow, Beck, Bey, Cuneo, Sison, Tobin, and Valeri (2004).

Agarwal, Shah, and Mannem (2011) proposed a system that generated questions from text using selective discourse connectives (*because*, *since*, *when*, *although*, *etc.*). The system had been evaluated manually by two evaluators. The system had scored 6.3 out of 8 for the QGSTEC2010 dataset and 5.8 out of 8 for the Wikipedia dataset. Liu, Calvo, Aditomo, and Pizzato (2012) presented an architecture for semi-automatic question generation to support academic writing. Their system extracted key-phrases from literature review articles. Then the conceptual graph structures were formed based on information, like section headings and section content in a Wikipedia article. Finally, the questions were generated based on this structure. Becker, Basu, and Vanderwende (2012) proposed an approach for generating quiz questions from the online text. Their problem of question generation had two parts: sentence selection and gap selection. They used their own implemented summarization technique, *SumBasic*, for selecting sentences. They presented a method to determine which gap in a sentence was used to ask a question by training a classifier. Mazidi and Nielsen (2014) presented a semantic approach to generate questions from science textbooks for self-study or assessment. The system obtained on an average 44% reduced error rate compare to Heilman and Smith (2010) and Lindberg et al.'s (Lindberg, Popowich, Nesbit, & Winne, 2013) systems. Majumder and Saha (2015) proposed a system to generate Multiple-Choice Questions (MCQs) from a text corpus on the sports domain. The system selected informative sentences using topic modeling and parse structure similarity. Then it found the answer-key from an informative sentence with the help of domain-specific and named-entity word. Finally, it generated distractors of the answer-key using a set of rules and a dictionary of name-entities. Liu et al. (2017) generated factual questions from a large Chinese corpus. They included linguistic knowledge in sentence simplification and used a set of rules for question generation such as abstract temporal adverb constraints, discourse relations in long Chinese sentences, and interrogative pronoun classification. In another approach, Liu, Rus, and Liu (2018) presented a mixed similarity strategy which generated distractors for Chinese multiple-choice questions with a statistical regression model. Das and Majumder (2017) proposed an approach for recognizing informative sentences from the text to generate factual open-cloze questions. Although the open cloze questions assess more productive-knowledge than the cloze-questions, examinees face the difficulty to guess the correct answer without alternative options. Therefore, they provided answer hints for the examinees to decrease the number of possible answers that make assessment easier. Table 1 has summarized the different types of questions found in the literature.

Table 1
The different types of questions found in the literature.

Existing work	Question type
Agarwal et al. (2011)	Wh-type, Example, Instance, Yes/No
Liu et al. (2012)	Wh-type
Becker et al. (2012)	Fill-in-the-blank/Cloze
Mazidi and Nielsen (2014)	Factoid Wh-type
Majumder and Saha (2015)	MCQ
Liu et al. (2017)	Factual Wh-type
Liu et al. (2018)	MCQ
Das and Majumder (2017)	Open Cloze

From the existing literature, we have found that many methods are used for generating questions. Deciding the most suitable approach for question generation in a particular application is very tough. Most of the articles generate questions from sentences, where the quality of questions depends on the extracted sentences. We have also noticed that, when questions are generated from a short text (e.g., a single document, headings, paragraph, or table content), keywords or key-phrases are suitable for generating questions. In this article, we have focused on generating subjective-questions from the syllabus with the help of multiword keywords to test the subject knowledge of students.

Evaluating subjective answers using computers reduces the time and effort of assessment in the next generation education system (Burrows, Gurevych, & Stein, 2015). Computer-assisted objective-type answer script evaluation is available (e.g., OMR or Optical Mark Recognition sheet), but assessing subjective answer script is in the initial stage of research due to the complex nature of the problem. We have discussed here some related works for evaluating subjective-answers.

Leacock and Chodorow (2003) developed a short answer grading system called *C-rater*, which deals with semantic information in the text. They used an approach of paraphrase recognition to grad answers. The *C-rater* was applied in two case studies for the experimental purpose: National Assessment for Educational Progress and a statewide assessment in Indiana. The method matched with human-graders about 84% of the time in both studies. Bin, Jun, Jian-Min, and Qiao-Ming (2008) applied the K-Nearest Neighbor (KNN) classifier for automatic essay scoring based on the well-developed text categorization model. The vector-space model was used to express each essay. They took words, phrases, and arguments as features of the essays, and the value of each vector was represented by the term frequency and inverse document frequency (TF–IDF) weight. The similarity of essays was calculated with cosine in the KNN algorithm. They achieved 76% average accuracy with the different methods of feature selection. Kakkonen, Myller, Sutinen, and Timonen (2008) proposed an automatic essay grading system that compares course materials and teacher's graded essays. They performed a set of experiments that used by the system for comparing three methods, Latent Semantic Analysis (LSA), Probabilistic LSA (PLSA), and Latent Dirichlet Allocation (LDA). They showed that the use of learning materials as training data for the grading model outperforms the KNN based grading system. Noorbehbahani and Kardan (2011) introduced a method for judging free-text answers of students using the Bilingual Evaluation Understudy (BLEU) algorithm. They

suggested a modified BLEU algorithm (M-BLEU) that was suitable for assessing the free-text answers. The M-BLEU identified the most similar reference answer to a student answer and calculated a similarity score to judge the answers provided by students. The evaluation results revealed that the suggested method achieved the highest correlation with human expert scores compared to other evaluation methods such as latent semantic analysis (LSA) and n-gram co-occurrence. Dhokrat et al. (2012) proposed an appraisal system for online subjective examination. They used a centralized file system for evaluating a student's answer, which included the reference material as well as the model answer for each question. A confidence factor of being positive was assigned when the required selective pattern of every correct answer matched with the model-answer. The accuracy of their system was found 70%. Islam and Hoque (2010) developed an essay grading system using generalized latent semantic analysis (GLSA). The system considered word-ordering in the sentences by introducing word n-gram for grading essays. The GLSA-based grading system overcame the limitations of LSA based system. The LSA based system had no way of detecting syntactic errors or when a few common words were missing in the sentence. In the LSA based system, students got full marks by writing an essay with only keywords. But their proposed system using GLSA overcame the above drawback of LSA based system. Ramachandran, Cheng, and Foltz (2015) presented a new technique for scoring short answer that used graphs for word-ordering to identify important patterns from human-provided rubric texts and top-scoring answers of the student. The method also used semantic metrics to determine groups of related words for alternative answers. They used two datasets for their experiment: Kaggle's short-answer dataset (Kaggle, 2012), and Mohler's short-answer dataset (Mohler, Bunescu, & Mihalcea, 2011). They showed that their method performed better than the best performing Kaggle entry and generalized it as a method to the Mohler dataset. Sakaguchi, Heilman, and Madnani (2015) used different sources of information for scoring short content-based assessment of student's response. They used a response-based method that used sparse features like word and character n-grams. They also used a reference-based approach that used a small number of features for the similarity between the response and information from the scoring guidelines. They found that their combined model outperformed over the result of 'response-based method' and 'reference-based method' when the training data is limited.

We have observed that the majority of the methods use a single similarity for assessment. The state-of-the-art similarity metrics, such as string similarity (Bin et al., 2008; Noorbehbahani & Kardan, 2011), semantic similarity (Islam & Hoque, 2010; Kakkonen et al., 2008), and keyword similarity (Dhokrat et al., 2012; Sakaguchi et al., 2015) are most popular in the literature. We found that individual similarity performs differently in a different mode of questions. This observation inspired us to use a fusion algorithm for the decision making process. In this paper, we have taken the advantages of all these similarities and proposed to use MCDM fusion for assessing student's subjective answers.

3. Proposed method

Subjective question has a long history of being used as a prime-tool for evaluating learner's knowledge and their understanding of a particular course or topic in a traditional education system (Jang, Sohn, & Cho, 2007). Towards research on an automated system for the subjective examination can reduce the effort of the traditional system (Alrehily, Siddiqui, & Buhari, 2018).

Our proposed system attempts to generate subjective questions and evaluates their answers automatically. It may be helpful to automate the subjective examination. The proposed method is subdivided into two modules: subjective-question generation from the course curriculum and answer assessment using a technique of decision making. To generate the subjective question, first, we have fetched the meaningful keywords from a set of course curriculum using Jaccard similarity coefficient (Jaccard, 1901) and then forming the question using selective subjective templates. Next, a set of model answers to each question is collected from subject experts for evaluating the student's response. We have considered the string similarity, semantic similarity, and keyword similarity (unigram, bigram, trigram, and noun) as the criterion of decision making. Finally, a weighted multi-criteria-decision-making (MCDM) approach is used to combine the scores and taking the final decision. Fig. 1 shows the overall framework of the automated subjective question generation and answer assessment system.

3.1. Subjective question generation

The proposed question generation task consists of two subtasks: extracting meaningful keywords and creating suitable subjective templates for question formation.

3.1.1. Keyword extraction

Different terms are used in the literature to represent the keyword in the document, such as key-phrase (Witten, Paynter, Frank, Gutwin, & Nevill-Manning, 2005), key-segment (Zhang, 2008), key-term (Lopez & Romary, 2010) or just keyword (Matsuo & Ishizuka, 2004). Keyword extraction is the task of identifying a set of keywords automatically from a document or corpus. We have used here the 'Jaccard similarity coefficient' (Tan et al., 2007) for extracting multiword keywords. It is a statistical measure used for comparing the similarity of finite sample sets. The Jaccard coefficient similarity between two sample sets A and B is defined by Eq. (1), where $0 \leq J(A,B) \leq 1$. If A and B are both empty, then we defined $J(A,B) = 1$.

$$J(A,B) = \frac{|A \bigcap B|}{|A \bigcup B|} = \frac{|A \bigcap B|}{|A| + |B| - |A \bigcap B|} \tag{1}$$

For multiword keyword extraction, the above equation is modified for co-occurrence similarity and re-written as Eq. (2).

$$J(x_1,x_2) = \frac{|x_1 \bigcap x_2|}{|x_1 \bigcup x_2|} = \frac{|x_1 \bigcap x_2|}{|x_1| + |x_2| - |x_1 \bigcap x_2|} \tag{2}$$

Here, $|x_1|$ is the frequency of first word (x_1), $|x_2|$ is the frequency of second word (x_2) and $|x_1 \bigcap x_2|$ is the joint frequency of two co-occurrence word-pair (x_1, x_2).

3.1.2. Subjective question templates

The subjective questions may be created in the following forms as reported in Dhokrat et al. (2012):

Define: Definition of a topic
Describe: Illustrate about the topic
Differentiate: Differences between two terms
Discuss: Details of a topic
Enumerate: Main points with brief explanations
Interpret: Analysis of the topic
Justify: Provide reasons that support the topic
Summarize: Main ideas in a brief
Trace: Main points in logical or chronological order

This paper has considered the following subjective templates that we have used to create subjective questions with the help of top-ranked keywords collected from the course curriculums.

Define <keyword><'.'>
What is <keyword><'?'>
What do you mean by <keyword><'?'>
Briefly describe <keyword><'.'>
Discuss <keyword><'.'>
Explain <keyword><'.'>
Write a short note on <keyword><'.'>
What are the differences between <keyword><'?'>

For explanation, let us consider two top-ranked trigram keywords *'threaded binary tree'* and *'linear data structure'* and also consider two

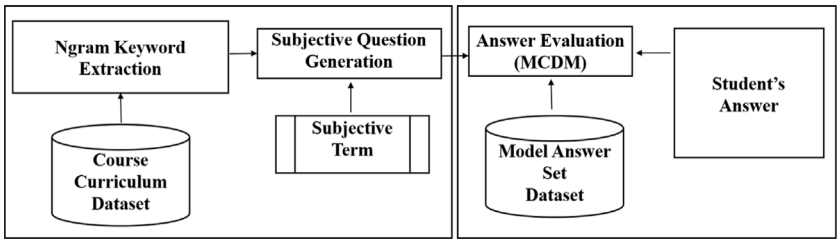

Fig. 1. Block diagram of subjective question generation and answer assessment system.

subjective templates: *'Define'* and *'What is'*. Therefore, the proposed system has produced the following two questions: *'Define threaded binary tree.'* and *'What is linear data structure?'*. The Algorithm 1 explains the steps of question generation.

Algorithm 1 Subjective Question Generation

Require: $S = \{s_1, s_2, ..., s_n\}$ and $ST = \{st_1, st_2, ..., st_n\}$ ▷ S is the set of course curriculums and ST is the set of subjective templates
Ensure: $Q = \{q_1, q_2, ..., q_n\}$ ▷ Q is the set of questions
1: $\forall S$ Calculate $f(x_i)$, $f(x_j)$ ▷ $f(x_i)$ is the frequency of word x_i, $f(x_j)$ is the frequency of next word x_j
2: Calculate $J(x_i, x_j)$ using Eq. (2)
3: Chose $KW = \{kw_1, kw_2, kw_3, ...kw_k\}$ ▷ KW is the set of top K keywords
4: Generate questions using subjective templates ST and keywords KW is a $m \times n$ matrix $\{ST, KW \rightarrow Q\}$
5: For generating questions using subjective term 'differentiate between kw_i and kw_j' where string similarity $StrSim(kw_i, kw_j) > \Theta$ ▷ kw_i and kw_j are any two keywords in the set KW, and Θ is threshold set to 0.7 heuristically

3.2. Assessment of subjective answers using similarity measure

The proposed system evaluates the answer to a subjective question. The main challenge of this task is the answer may be written in various ways using the different forms of sentences (active or passive) and synonyms or word forms, etc. For example, the sentence, *'A purchased a car from B'*, may be written as *'A bought a car from B'* or *'B sold a car to A'*. Therefore, to evaluate a given answer, we have to consider various forms of similarity. In the proposed method, first, we have checked the similarity of a student response with the model answer set using different measures. Then the score is calculated using the MCDM-based fusion technique. We have performed a few preprocessing tasks before applying various similarity measures. A online grammar checker plugin (Naber, 2003) has been used for track down the grammar, spelling, and punctuation errors. Stanford Parser and dependency manual (De Marneffe & Manning, 2008) are conferred with to ensure the structure or formation of the sentence in the student's response. In the next subsections, we have explained the various similarity measures used for evaluating the answers that are written using a complete sentence.

3.2.1. String similarity metric

A string similarity metric is used to calculate the distance between two given text strings. The most widely used technique for such task is Levenshtein distance (LD) (Navarro, 2001). It works on two input strings; returns a number equivalent to the number of substitutions, deletions, and insertions needed to transform one input string into another. It has expanded to include phonetic, token, grammatical, and character-based approaches of statistical comparisons. The Algorithm 2 explains the string similarly α of a student's answer sa to model-answer set MA.

Algorithm 2 String Similarity (α)

Require: $MA = \{ma_1, ma_2, ..., ma_p\}$ ▷ MA is the set of model answers
Ensure: α ▷ α is the string similarity of a student's answer sa
1: $Max = 0$
2: **for all** $ma \in MA$ **do**
3: Calculate $LD(ma, sa)$
4: **if** $Max < LD(ma, sa)$ **then**
5: $Max = LD(ma, sa)$
6: **end if**
7: **end for**
8: **return** Max

3.2.2. Semantic similarity metric

Semantic similarity is a metric that uses a set of documents or sentences to measure the distance. The method uses the likeness of their meaning or semantic content in contrast with string similarity, which calculates their syntactical representation. Semantic similarity is estimated by topological similarity using an ontology that calculates the distance between terms or concepts (Harispe, Ranwez, Janaqi, & Montmain, 2015). Semantic relatedness between words or sentences is calculated using statistical measures such as a vector space model (Feng, Bagheri, Ensan, & Jovanovic, 2017). For our experiments, a well-known approach latent semantic analysis (LSA) (Aouicha, Taieb, & Hamadou, 2018) is used to check the similarity of a given answer with the model-answer set. The Algorithm 3 explains the semantic similarly β of a student's response sa to model answer set MA.

Algorithm 3 Semantic Similarity (β)

Require: $MA = \{ma_1, ma_2, ..., ma_p\}$ ▷ MA is the set of model answers
Ensure: β ▷ β is the semantic similarity of a student's answer sa
1: $Max = 0$
2: **for all** $ma \in MA$ **do**
3: Calculate $LSA(ma, sa)$
4: **if** $Max < LSA(ma, sa)$ **then**
5: $Max = LSA(ma, sa)$
6: **end if**
7: **end for**
8: **return** Max

3.2.3. Keyword similarity metric

Keyword similarity is a metric that measures the keyword matching with a given answer to the model answer set. In our experiment, different types of keyword matching measures are taken. We have extracted unigram, bigram, trigram keywords, and noun-phrases by eliminating the stopwords and punctuation marks from a student's answer. Suppose, n_1 is a unigram-keyword that has one word, n_2 is a bigram-keyword that contains two co-occurrence words, and n_3 is a trigram-keyword consist of three consecutive words. Let us consider, D is the number of total keywords ($n_1 + n_2 + n_3$) and ϕ is the number of nouns of student's answer matched with the model answer set MA.

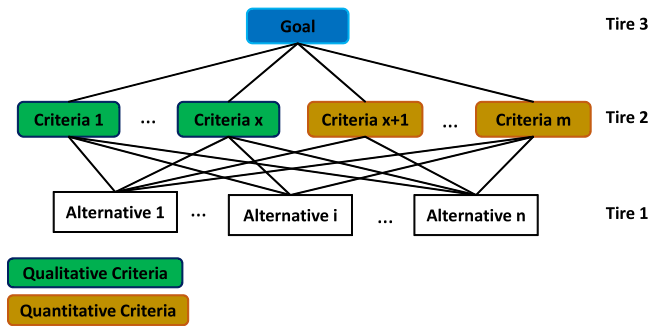

Fig. 2. Block diagram of typical MCDM.

Then the keyword similarity γ is normalized using Eq. (3).

$$\gamma = \frac{\sum_{n=1}^{3}(D) + \phi}{N}, \ where \ N \ is \ \max(\sum_{n=1}^{3}(D) + \phi) \tag{3}$$

3.3. Subjective answer evaluation and scoring

The final step of the method is to evaluate and perform scoring. To achieve this, we have used the method of Multi criterion Decision-Making (MCDM). The MCDM is gaining importance as a potential tool for analyzing complex real-problems due to its inherent ability to judge different alternatives on various criteria for possible selection of the best/suitable alternative(s). Fig. 2 depicts the MCDM technique for possible alternative selection. We have used a variation of MCDM-based fusion for scoring a given answer.

The weighted sum model (WSM) is one of the earliest and probably the most widely used method in MCDM (Triantaphyllou, 2000). The weighted product model (WPM) can be admitted as an alteration of the WSM. The analytic hierarchy process (AHP) is a later development, and it has recently become popular (Saaty, 2008). Recently modification of AHP is considered more consistent than the original method. Some other widely used approaches are the ELECTRE and the TOPSIS (Zanakis, Solomon, Wishart, & Dublish, 1998).

The WSM is the well-known and simplest MCDM method for evaluating a number of alternatives in terms of a number of decision criteria. It is important to state that the WSM is applicable when all data are expressed in exactly the same unit. In this paper, we have considered first a typical WSM model for fusion. Then we have compared with other methods to test the accuracy of other MCDM methods in Experiment #3.

Let us consider a typical MCDM problem based on n alternatives and m decision criteria. Next, assume that all the criteria are beneficial, and the higher the value is, the better it is. Here, w_j indicates the relative weight of importance of the criterion C_j and a_{ij} is the performance value of alternative A_i when it is evaluated in terms of criterion C_j. Then, the total (all criteria are simultaneous) importance of alternative A_i, denoted as φ is defined in Eq. (4).

$$\varphi = \sum_{j=1}^{m} w_j a_{ij}, \quad \text{where} \quad i = 1, 2, 3, \ldots, n \tag{4}$$

For example, let the scores (α, β, γ) of an answer are 0.767123288, 1.0, and 0.545454545 respectively and the weights are set as 0.2, 0.2, and 0.6. The final WSM score (φ) is calculated in Eq. (5).

$$\varphi = 0.767123288 \times 0.2 + 1 \times 0.2 + 0.545454545 \times 0.6 = 0.680697385 \tag{5}$$

The selection of weights is one of the important steps in the weighted multi-criteria-decision-making process used in this article. The general practice for selecting suitable weight is to depend on expert (Odu, 2019), or use a trial-and-error method. In experiment #2, we have observed that the accuracy is high when the weights setting are {0.2, 0.2, 0.6}.

The WSM is the most widely used method of MCDM that calculates the individual value of each alternative for different decision criteria (Eq. (4)). The WPM is used to overcome some of its weaknesses. An advantage of this method is that instead of the individual values, it can use relative ones. In our experiment, we have evaluated the answers of students, not required to rank the students according to their answers. Therefore, we have taken the individual values of alternatives, not taken the relative values among the alternatives for ranking. The WPM is also applicable in this case. We have stated the WPM in Eq. (6) for calculating the individual value of an alternative A_i.

$$\psi = \Pi_{j=1}^{m}(a_{ij})^{w_j}, \quad \text{where} \quad i = 1, 2, 3, \ldots, n \tag{6}$$

The TOPSIS method is mainly used to get an optimal solution, which is farthest from the negative ideal solution and closest to the positive ideal solution (Assari, Mahesh, & Assari, 2012). We have modified this equation to get the individual score of an alternative using Eq. (7) and Eq. (8) where the positive ideal solution is set by 1, and the negative ideal solution is set by 0. The ω_j^+ is the separation of every alternative from the positive ideal solution, and the ω_j^- is the separation of every alternative from the negative ideal solution.

$$\omega_j^+ = \sqrt{\sum_{j=1}^{m}(a_{ij} - 1)^2} \tag{7}$$

$$\omega_j^- = \sqrt{\sum_{j=1}^{m}(a_{ij} - 0)^2} \tag{8}$$

The closeness to the ideal solution of each alternative is calculated from Eq. (9).

$$\tau = \frac{\omega_j^-}{\omega_j^- + \omega_j^+} \tag{9}$$

The Algorithm 4 summarizes the answer assessment technique using an MCDM-based method.

Algorithm 4 Answer Assessment

Require: $SA = \{sa_1, sa_2, ..., sa_n\}$ ▷ SA is the set of answers given by the students
Ensure: $R = \{r_1, r_2, ..., r_n\}$ ▷ R is the set of assessment results, either 1 for correct answer or 0 for wrong answer

```
Calculate α          ▷ α denotes the string similarity of Section 3.2.1
Calculate β          ▷ β denotes the semantic similarity of Section 3.2.2
Calculate γ          ▷ γ denotes the keyword similarity of Section 3.2.3
Then, ψ = MCDM (α, β, γ)          ▷ ψ calculate the score of answer assessment
if ψ ≥0.5 then
    R ← 1
else
    R ← 0
end if
```

4. Experiments and results

This section reports the performance of the proposed question-generation and answer-assessment system. First, we have discussed the proposed dataset and preprocessing technique. Then we have explained the experiments and results. Finally, we have discussed the results of several case studies. The accuracy is estimated using the confusion matrix in Eq. (10).

$$Accuracy = \frac{(TP + TN)}{TP + FP + FN + TN} \tag{10}$$

Where TP=True Positive, TN=True Negative, FP=False Positive and FN=False Negative.

4.1. Dataset

In our study, we have not found any suitable public dataset for evaluating the proposed method. Different authors proposed different dataset to evaluate their free-text answers that are different in nature and scoring scale are also different (Roy, Narahari, & Deshmukh, 2015). Therefore, we have prepared a dataset called DBSYLLABI to evaluate the system. The dataset contains three different courses of computer science; *'Data Structure'*, *'Cryptography'* and *'Operating System'*. Each course consists of a set of 35 different course curriculums. The course curriculums are collected from books, different colleges, and university websites. Then this dataset is fed to the system for generating questions. The model-answers are collected by subject experts or books/online resources. To examine our proposed assessment system, we have collected 25 model answers for each question to evaluate 100 answers of students in the dataset.

4.2. Preprocessing

The corpus DBSYLLABI is preprocessed in such a way that the frequency of word and co-occurrence word-pair are easily counted from the course curriculums and reduces the effort of extracting meaningful n-grams. To extract the co-occurrence word-pair, we have split the sentences where punctuation and stop-words have occurred, and then they are removed from the corpus. It is taken into account that if any punctuation or stopword is present in between any two consecutive words, then they are not considered as a co-occurring word-pair. Then n-grams ($2 \leq n \leq 3$) are extracted from each line and count the frequency of $f(w_i, w_j)$ and $f(w_i, w_j, w_k)$ for each unique n-grams. we have also extracted n consecutive nouns (NNP/NN) ($1 \leq n \leq 3$) from the each line using part-of-speech (POS) tagger (Schmid, 1994).

4.3. Experiments

Experiment #1: In the first experiment, we have evaluated the capability of generating relevant questions from course curriculums. After generating the questions, four experts are asked to mark relevant/irrelevant questions, and these are considered as the ground truths for each expert. The average accuracy is defined by taking average consideration of the system according to expert marking (relevant considered as correct and irrelevant considered as incorrect). Fig. 3 presents the average accuracy of experts for generating relevant questions is accumulated by varying the number of course curriculums considering a fixed number of keywords; 25 is taken in our case..

Fig. 4 presents the average performance evaluation suggested by experts for generating questions by varying the number of keywords considering a fixed number of course curriculums; 35 is taken in our case. The method is also validated by taking experts marking of relevant/irrelevant questions. It is observed that the model generates more irrelevant questions when the number of top-ranked keywords (let K) is increased. Fig. 4 shows non-linear behavior, because the accuracy of question generation based on top-K keywords depends on the number of important keywords present in the corpus. If the corpus contains less than K number of important keywords, the method tends to choose wrong words as keywords; hence, the generated questions based on the keywords are also wrong. We conclude that optimum K purely depends on the content of the dataset.

Experiment #2: In the second experiment, we have evaluated the performance of the proposed answer-assessment technique. First, the student answers are evaluated by 4 experts and marked as correct/incorrect. The average accuracy is the average system accuracy with respect to all 4 experts. In the case of MCDM-based fusion, we have experimented to understand the effect of the weights of each criterion. The accuracy varying the weights is reported in Table 2. Table 2 shows that the accuracy is better, when the weights setting is {0.2, 0.2, 0.6}.

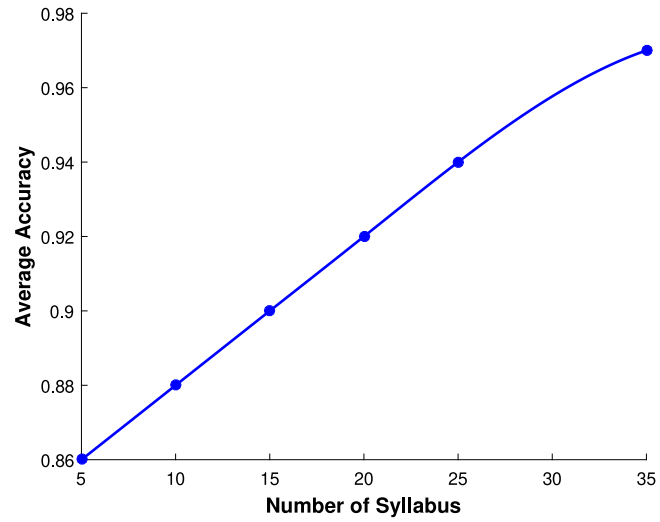

Fig. 3. Accuracy of question generation by varying the number of course curriculums when the number of keywords is fixed.

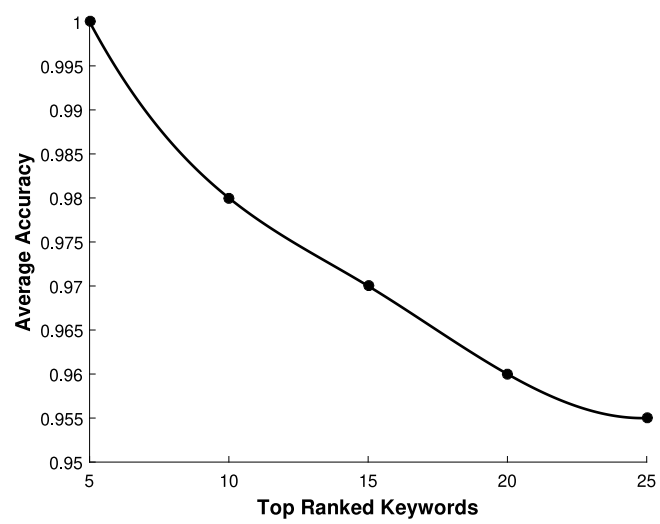

Fig. 4. Accuracy of question generation by varying the number of keywords when the number of course curriculums is fixed.

Table 2
Accuracy of subjective answer evaluation.

Weightage of string similarity	Weightage of semantic similarity	Weightage of keyword similarity	Accuracy of answer evaluation
0.33	0.33	0.33	0.695
0.8	0.1	0.1	0.565
0.1	0.8	0.1	0.62
0.1	0.1	0.8	0.74
0.2	0.1	0.7	0.76
0.2	0.2	0.6	0.79

Next, the model is validated with the ground truth by taking the weights as {0.2, 0.2, 0.6} and the average accuracy is reported in Fig. 5. We have calculated the answer score and check whether the answer is correct or not based on the threshold of 0.5. We have observed that different similarity metrics use different forms of similarity. For example, the string similarity outperforms when the questions like *"Define something"*, because the definition of some well-known phenomena is presented using a similar linguistic term. But, questions like *"Explain something"*, the string similarity may not be the best choice, as explanations can be different for different students. A similarity metric

Table 3
The 10 different case study values of the answer evaluation for the question 'what is data structure?'.

Case	String similarity (α)	Semantic similarity (β)	Keyword similarity (γ)	WSM score (φ)	Expert	Proposed method
	Weight=0.2	Weight=0.2	Weight=0.6			
Case #1	0.767123288	1	0.545454545	0.680697385	correct	correct
Case #2	0.48427673	0.746003847	0.545454545	0.573328843	correct	correct
Case #3	0.403846154	0.9258201	0.454545455	0.538660523	incorrect	correct
Case #4	0.423076923	0.918558654	0.090909091	0.32287257	incorrect	incorrect
Case #5	0.504587156	0.774596669	0.363636364	0.474018583	correct	incorrect
Case #6	0.593333333	0.83567165	0.909090909	0.831255542	correct	correct
Case #7	0.455445545	0.629940788	0.272727273	0.38071363	incorrect	incorrect
Case #8	0.458715596	0.668153105	0.545454545	0.552646467	correct	correct
Case #9	0.412587413	0.632455532	0.272727273	0.372644953	incorrect	incorrect
Case #10	0.359183673	0.824786099	0.727272727	0.673157591	correct	correct

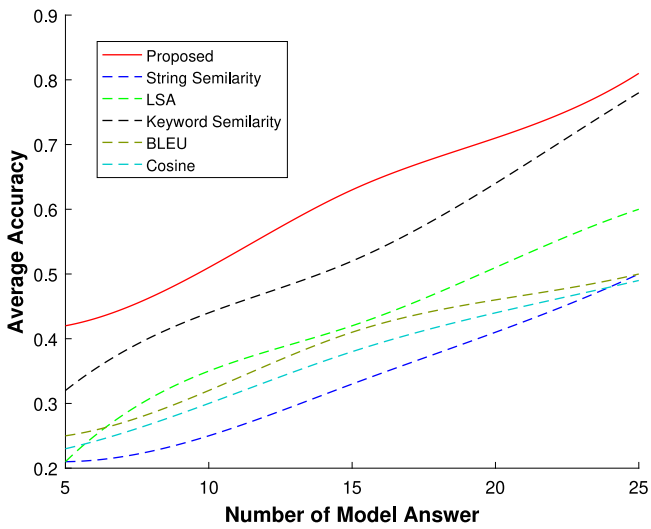

Fig. 5. Accuracy of the answer-assessment varying the number of model-answers. It is observed that the accuracy is as high as 79% using the proposed method when we have added 25 model-answers for assessment.

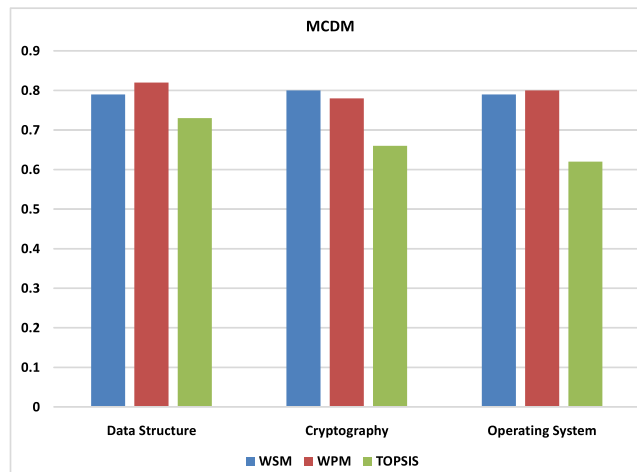

Fig. 6. Accuracy of answer assessment using different MCDM method.

like BELU can be a good option here. On the other side, when we have a large number of model answers and have a large set of possible linguistic ways of writing answers to a question. Then, string similarity can outperform BELU (as seen in our case). But, the overall performance again will better if we use multiple similarity metrics together. This observation leads us to use multiple-similarity metrics together. Fig. 5 presents accuracy varying different methods and number of model questions. We observed that the majority of the similarity metrics are linear to the number of the model answers. The figure concludes that the fusion of multiple similarity metrics can outperform individual similarity metrics. Although we can say that the final accuracy using the proposed fused method partially depends on the dataset, a suitable selection of weight for a different dataset can outperform individual metrics.

Case Study: In this section, we have presented the discussion of experimental results obtained from the MCDM using several case studies. The case studies are selected based on their variation. We have considered the 10 cases for showing the variation of different similarity measures and taken the MCDM score for evaluation. The sample answers are evaluated for the question 'What is data structure?' Table 3 summarizes the cases.

Case #1: In this case, the string similarity and semantic similarity are high, whereas keyword similarity and WSM score are average. According to the experts, the answer is correct, and the proposed method also marked the answer as correct.

Case #2: The semantic similarity is high, but all other scores are average. The proposed method agrees with the experts and marks as correct.

Case #3: In this case, the semantic similarity is very high, string similarity is low, keyword similarity and WSM scores are average. In this case, the proposed method fails and classifies the answer as correct, whereas experts evaluate the answer as wrong.

Case #4: In case #4, the semantic similarity is very high, and all other scores are very low. The proposed method agrees with the experts and classifies the answer as wrong.

Case #5: In this case, the semantic similarity is high, string similarity is average, and other scores are low. The proposed method fails and classifies the answer as wrong, whereas the experts evaluate it as correct.

Case #6: The semantic similarity and keyword similarity are high. The string similarity is average. The method agrees with the experts and classifies the answer as correct.

Case #7: In case #7, there exists average semantic similarity and string similarity. The keyword similarity and WSM scores are low. The method agrees with the experts and evaluates the answer as wrong.

Case #8: In this case, all of the scores are average. The method evaluates the answer as correct similar to the expert's evaluations.

Case #9: In case #9, the semantic similarity and the string similarity are average, but the other two scores are very low. The method successfully detects the answer as wrong.

Case #10: In this case, the string similarity is low, the WSM score is average, and the other two scores are high. The proposed method successfully classifies the answer as correct.

Experiment #3: A comparative study of the WSM with other MCDM methods is also admitted in the third experiment. Fig. 6 shows the comparative study of answer assessment using different MCDM methods.

5. Discussion of the results

We have noted from the literature that the majority of the articles used the string-similarity, semantic similarity, and keyword similarity in different ways to assess the written responses of students. In this paper, we have proposed to use the MCDM-based fusion technique to take advantage of all these similarities for subjective assessment. We have initially used the WSM method for fusion. Then we have compared it with other methods to test the accuracy of other MCDM methods. We have conducted three experiments to analyze the effectiveness of the proposed method for question generation and answer assessment. First, we have analyzed the effectiveness of the keyword-based question generation method. Fig. 3 presents the results. We have noted that the average accuracy (according to 4 experts) almost linear with the number of syllabuses considered for keyword extraction. We have also noted that the accuracy increase only ∼12% when the number of the syllabuses is increased 5 to 35. Fig. 4 presents the average accuracy of the question generation method, varying the number of keywords. We have noted that the inclusion of a higher number of keywords from a fixed syllabus sometimes decreases the accuracy when wrong keywords are included. We have noted that the accuracy decreases only ∼5% when the number of keywords changes from 5 to 25. In Experiment #2, we have critically analyzed the effectiveness of the proposed weighted sum method for answer evaluation. We have heuristically checked the weight settings and found that a setup of $\{0.2, 0.2, 0.6\}$ is suitable for our dataset. We have noticed that different methods outperform in different datasets, such as WPM is better for our *'data-structure'* dataset, WSM for *'cryptography'*, and WPM for the dataset of *'operating system'*. The MCDM method may also be applicable when ranking the students according to their correct answers. In Experiment #3, we have compared the proposed method with the state-of-the-art. We have observed that almost the majority of the similarity metrics are linear to the number of the model answers. Fig. 6 concludes that the fusion of multiple similarity metrics can outperform individual similarity metrics. Although we can say that the final accuracy using the proposed fused method partially depends on the dataset, a suitable selection of weight for a different dataset can outperform the accuracy of the individual metric.

6. Conclusion

Automatic subjective question generation and answer evaluation are challenging areas in NLP and the education field of research. Our proposed work introduces an automated question generation system from the course curriculum with an evaluation system when the answer to the question is written in a single sentence. We have fetched appropriate keywords from the course curriculum to generate questions by using subjective templates. The MCDM-based approach is proposed here to evaluates the answers. The string similarity, semantic similarity, and keyword similarity are considered as multi-criteria to evaluate answers. The overall performance of our system highlights that it can be helpful for subjective assessment.

The main limitations of the proposed method are: (a) the selection of the accurate number of keywords is sensitive to the dataset. We observed that in some cases increasing the number of keywords improves the performance (when the set of curriculum contains a large number of keywords); but, this is not true for all cases (when the set of curriculum contains a limited number of keywords). (b) The selection of the weight setting is also sensitive to the dataset. The same weight setting may not be suitable for all kinds of datasets.

The extension of the work may be deemed as MCDM fusion to evaluate answers written in multiple sentences. We are also interested to find suitable weight settings using optimization techniques. The inclusion of other similarity measures, such as knowledge-based similarity and deep similarity will be considered in the future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors. Informed consent: Informed consent was obtained from all individual participants included in the study.

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